# BAGAS ADJIE PAMUNGKAS

**CURRICULUM VITAE**

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## 1. PERSONAL DETAILS

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|---|---|
| **Full Name** | : Bagas Adjie Pamungkas |
| **Student ID (NIM)** | : 24/544718/TK/60547 |
| **Study Program** | : Bachelor of Information Engineering (Program Studi Teknologi Informasi) |
| **Department & Faculty** | : Dept. of Electrical and Information Engineering (DTETI), Faculty of Engineering, UGM |
| **Domicile** | : Sleman, Daerah Istimewa Yogyakarta, Indonesia |
| **Languages** | : Indonesian (Native), English (Professional Working Proficiency) |

## 2. PERSONAL SUMMARY

Information Engineering undergraduate at Universitas Gadjah Mada with project experience in data science, machine learning, and software development. Experienced in building end-to-end data preprocessing pipelines, predictive models, and scalable applications, alongside active leadership in organizing department-level technical training workshops and national programming competitions.

## 3. FORMAL EDUCATION HISTORY

**Universitas Gadjah Mada (UGM)** – Sleman, D.I. Yogyakarta **2024 – Expected 2028**
*Bachelor of Engineering in Information Engineering (DTETI FT UGM)* **Cumulative GPA: 3.38 / 4.00**

- **Core Focus Areas:** Artificial Intelligence, Machine Learning, Data Science, Software Engineering, and Cloud Computing.

**SMA Negeri 10 Yogyakarta** – Yogyakarta, D.I. Yogyakarta **2021 – 2024**
*Senior High School – Mathematics and Natural Sciences (MIPA)*

- Active member of the school environmental and sustainability organization (*Adiwiyata SMAN 10 Yogyakarta*).

**SMP Negeri 15 Yogyakarta** – Yogyakarta, D.I. Yogyakarta **2018 – 2021**
*Junior High School Education*

**SD Negeri Jetisharjo Yogyakarta** – Yogyakarta, D.I. Yogyakarta **2012 – 2018**
*Elementary School Education*

## 4. ORGANIZATIONAL & COMMITTEE EXPERIENCE

**KMTETI FT UGM (Keluarga Mahasiswa Teknik Elektro dan Teknologi Informasi)** **Nov 2024 – Present**
*Staff of Workshop Division*

- Facilitated technical training programs, coding bootcamps, and hands-on engineering workshops (Data Science, Web Development, and Microcontroller Systems) for Electrical and Information Engineering students.
- **Organizing Committee – Pelatihan Web Development 2 (Even Semester):** Spearheaded the planning and execution of the even-semester Web Development 2 training work program (*proker*), managing curriculum design, speaker/mentor coordination, and hands-on coding sessions.
- **Competition Committee – TETI Programming Week (Data Science Branch):** Managed the Data Science competition track work program (*proker*), overseeing problem case formulation, dataset curation, evaluation setup, and participant technical assistance.

**Find IT! (Future Information Technology) UGM 2025** **Oct 2024 – May 2025**
*Staff of Equipment, Logistics, and Consumption (Perlogkom) Division*

- Coordinated end-to-end equipment procurement, venue readiness, and catering operations across national IT competition tracks (Hackathon, Competitive Programming, Capture The Flag, and Data Analytics).
- Co-designed technical fabrication and hardware specifications for the *Capsule of Innovation* opening ceremony installation.

**Technocorner UGM 2025** **Oct 2024 – May 2025**
*Staff of Equipment, Logistics, and Consumption Division*

- Managed technical equipment readiness, inventory tracking, venue setup, and on-site logistics for staff and participants throughout the preparation and execution phases of DTETI FT UGM's national technology event.

**Adiwiyata SMA Negeri 10 Yogyakarta** **Jan 2022 – Jan 2023**
*Staff of Adiwiyata (Environmental & Sustainability Organization)*

- Organized school-wide waste sorting and processing initiatives, green-space spatial arrangement, and routine plant conservation programs.

## 5. HONORS & AWARDS

- **First Runner Up (2nd Place) – Final Project Bootcamp TETI Mobile Development Jun – Jul 2026**
  *Divisi Workshop KMTETI FT UGM (Team: MobDev-Kelompok 3)* – Awarded 2nd place for developing an end-to-end mobile application final project.
- **First Runner Up (2nd Place) – Final Project Bootcamp TETI Artificial Intelligence Jul 2025**
  *Divisi Workshop KMTETI FT UGM (Team: Remaja Masjid)* – Awarded 2nd place for building and presenting an applied AI/ML final project.

## 6. TECHNICAL & RESEARCH PROJECTS

**Customer Service Ticket Triage & RAG Copilot** GitHub

- Fine-tuned multi-task IndoBERT (indobert-base-p1) with PEFT/LoRA (2.41% trainable params) and GroupShuffleSplit (user-level) leakage checks across 1,200 support tickets for 4-class category and Focal-Loss urgency routing, lifting test Category Macro-F1 from 0.7713 to 0.9818, Urgency Macro-F1 from 0.6259 to 0.9644, and Critical ticket recall from 0.5357 to 1.0000 over the TF-IDF+LightGBM baseline.
- Added OOF temperature scaling (T=0.60) and $F_\beta$-tuned Critical threshold (0.495) to keep calibrated confidences without degrading test metrics, plus a rule-based regex PII masker and entity extractor (Exact-Match F1 1.0000 on the synthetic set), and a hybrid BM25+E5 RAG retriever (Recall@5 1.0, MRR@5 0.997).

**KlaimKilat AI – Omnichannel Reimbursement & Audit Agent** GitHub

- Built a stateful LangGraph agent that ingests receipts from web scanner, Telegram, and WhatsApp, runs vision extraction, hybrid-RAG SOP matching, and quota/audit tool calls to auto-approve, auto-reject, or escalate claims via human-in-the-loop interrupts.
- Engineered a hybrid policy retriever (pgvector dense + Postgres full-text search fused with Reciprocal Rank Fusion, k=60) over table-aware SOP chunks, reaching Recall@4 1.0 and MRR 1.0 across 40 benchmark scenarios, served through FastAPI + Redis with per-node latency/token-cost telemetry.

**TransitSentinel – Cashless Transit Anomaly Detection & Settlement Audit** GitHub

- Engineered a real-time anomaly detection and settlement audit pipeline over 120,000 transit tap transactions across 45 stations and 120 readers with 28 causal spatio-temporal features and zero temporal leakage.
- Built a hybrid 3-tier engine combining sub-0.1ms deterministic guardrails, PyTorch Residual VAE, and Isolation Forest + Calibrated LightGBM stacking with TreeSHAP and cost-sensitive routing ($\tau^* = 0.41$), achieving 0.9868 PR-AUC, 100% double-tap auto-refund precision, 98.0% net cost savings, and sub-25ms p95 latency on FastAPI, plus daily settlement reconciliation auditing via Robust MAD Z-Score.

**Bengawan Solo River Water Level Forecasting** GitHub

- Engineered 187 leakage-free features across 30 stations (84,396 train and 888,480 environmental records), combining FAO-56 VPD, soil-moisture runoff, 30/60/90-day reservoir lags, and HydroRIVERS/IDW spatial weights.
- Built a 7-member Huber ensemble (LightGBM, XGBoost, CatBoost, per-station Ridge Fourier) with 4-fold rolling-origin CV and causal decaying-residual post-processing, lowering mean 4-fold validation MAE to 0.536 (RMSE 1.394), a 8.2% reduction over the diversified baseline.

**Pothole Semantic Segmentation Pipeline** GitHub

- Built a 5-fold StratifiedGroupKFold ensemble combining UNet++ (EfficientNet-B4 scSE), SegFormer (mit-b2), and DeepLabV3+ (SE-ResNeXt50) with 5-branch augmentation (1,990 samples/epoch) and RAM caching.
- Implemented hybrid Lovasz-Hinge loss warmup, WarmupModelEMA, multi-scale TTA (0.75x, 1.0x, 1.25x), and hysteresis post-processing with RLE export, achieving 0.8327 Validation Dice (0.7336 mIoU).

**Hierarchical Waste Classification with SigLIP & Satria Data 2026** GitHub

- Extracted frozen 1,152-dim embeddings using google/siglip-so400m-patch14-384 across a cleaned 28,965-image dataset (20,275 train / 4,345 val / 4,345 test) spanning 3 primary categories and a coarse-to-fine taxonomy of 8 trained sub-labels.
- Built a cross-family ensemble (multi-seed residual MLP + cosine kNN + RBF SVM) with effective class weighting, label smoothing, and embedding-space MixUp, reaching 0.9956 validation Macro-F1 (0.9936 test) on primary categories with 0.9958 validation B3 recall, and 0.9591 Macro-F1 on 8 human-verified sub-labels (B3 sub-label recall $\geq$ 0.97, precision 1.00).

**The 15-Minute City Profiler – Urban Spatial Accessibility System** GitHub

- Profiled DKI Jakarta walkability over 18,782 grid-aggregated OpenStreetMap facilities across 5 SDG-11 service categories (education, transit, green space, health, economy) via OSMnx, implementing a 25x25 spatial grid, Normalized Shannon Entropy diversity indexing, neighbor-weighted spatial smoothing ($\alpha$=0.4), and K-Means++ from scratch in pure NumPy (optimal K=2, Silhouette 0.4786; 31 ideal zones vs 26 standard), mapping 15-minute-city area coverage per administrative city with Folium.

**DKI Jakarta Flood Risk Data Engineering & PCA** GitHub

- Integrated 7 multi-domain open datasets (876,447 Dinas SDA telemetry records, BPBD flood logs, BMKG daily weather, BPS monthly climate, Dukcapil demographics, administrative polygons, and 10m Sentinel-2 Dynamic

World GEE rasters) across 267 Jakarta sub-districts with datum-free feature engineering and leak-free PCA (73.65% cumulative variance, 3 Kaiser-Guttman PCs).

**Face Anti-Spoofing Detection & NaviGreen Eco-Routing** GitHub

- Developed a 5-fold ConvNeXt-Tiny face anti-spoofing pipeline over 6 classes (realperson, screen, printed, mask, mannequin, unknown) with SigLIP mislabel detection, MD5 cross-class leakage hashing, per-fold class weighting, Focal Loss, Optuna tuning, and Grad-CAM explainability.
- Co-designed NaviGreen for mobile eco-routing and carbon scoring that penalizes road grade and aggressive driving to reduce fuel consumption and carbon emissions.

**Rekberin – Game Account Marketplace with Escrow Handover** GitHub

- Co-developed (team of 5) a full-stack game-account marketplace in Next.js/TypeScript with PostgreSQL (Prisma), NextAuth, and real-time buyer-seller chat, offer negotiation, and a two-hour escrow-style account handover confirmation flow.
- Integrated Midtrans QRIS payments with a background worker and scheduled job (`/api/jobs/handover-expiry`) that enforces handover deadlines, dispute reporting with admin mediation, and simulated payout/refund settlement.

## 7. CERTIFICATIONS, TRAINING & TECHNICAL SKILLS

- **AWS Academy Graduate – Cloud Web Application Builder Training Badge** (12 Hours; Issued Oct 3, 2026 | Credly Badge)
- **DataCamp Statements of Accomplishment (Python Track):** *Dimensionality Reduction in Python* (#50,052,989; 4 Hours; Sep 24, 2026); *Exploratory Data Analysis in Python* (#49,969,581; 4 Hours; Sep 19, 2026); *Cleaning Data in Python* (#49,881,206; 4 Hours; Sep 15, 2026).
- **Data Science Training – Workshop Division, KMTETI FT UGM** (Completed Sep 3 – Nov 16, 2025)
- **Additional Training & Workshops:** AWS Academy (Microservices & CI/CD Pipeline Builder); Kiro x Agentic AI Workshop (DTETI UGM); Kaggle 5-Day AI Agents Intensive Course with Google.

**Programming :** Python, SQL, C++, C, JavaScript, HTML, CSS.

**AI/ML, NLP & Computer Vision:** PyTorch, Hugging Face, PEFT/LoRA, RAG, IndoBERT, Variational Autoencoder (VAE), Isolation Forest, TreeSHAP, UNet++, SegFormer, DeepLabV3+, ConvNeXt, SigLIP, OpenCV

**Data Science, Geospatial & Cloud:** Pandas, NumPy, SciPy, Scikit-learn, LightGBM, XGBoost, CatBoost, Optuna, GeoPandas, OSMnx, GEE, FastAPI, Redis, Streamlit, AWS, Git, Docker